

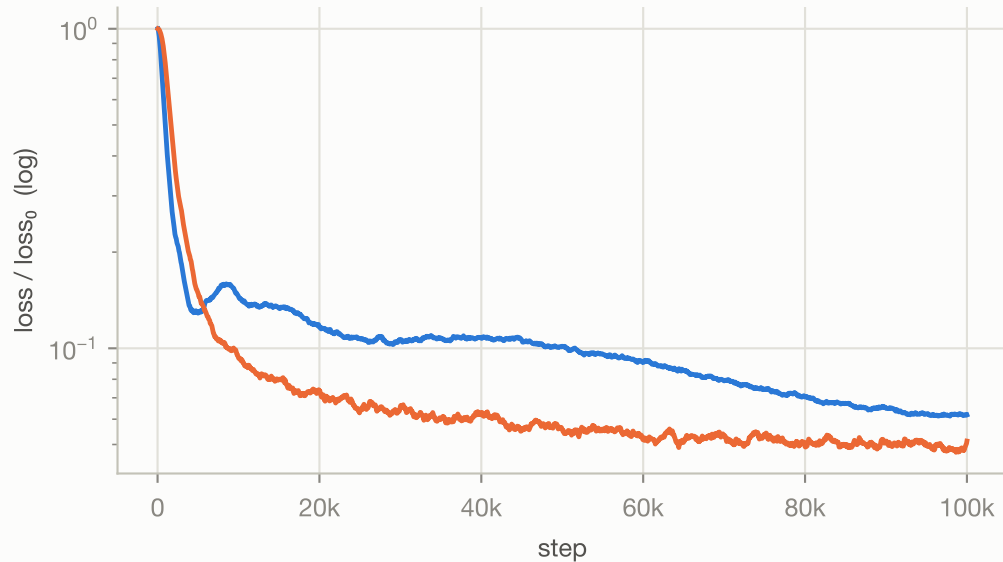
vJEPA vs vMAE — Rayleigh-Bénard

4ch · 512×128 · buoyancy-driven convection

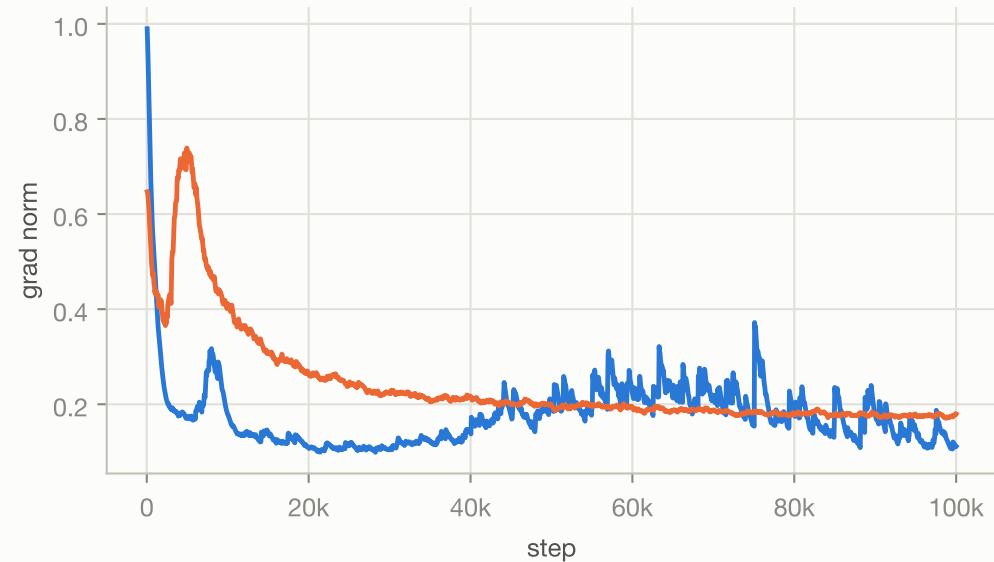
100,000 steps each · identical ViT-S encoder, tube masking @0.9 — wall-clock: JEPA 8.7h / MAE 3.2h — final loss (raw): JEPA 0.032 / MAE 0.066

— JEPA (latent prediction)
— MAE (pixel reconstruction)

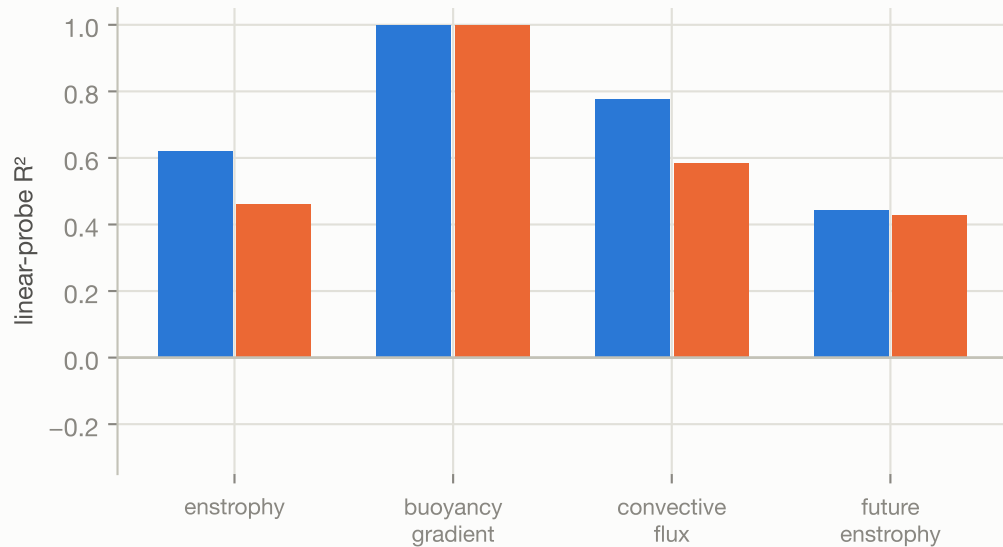
Training loss, relative to step 50



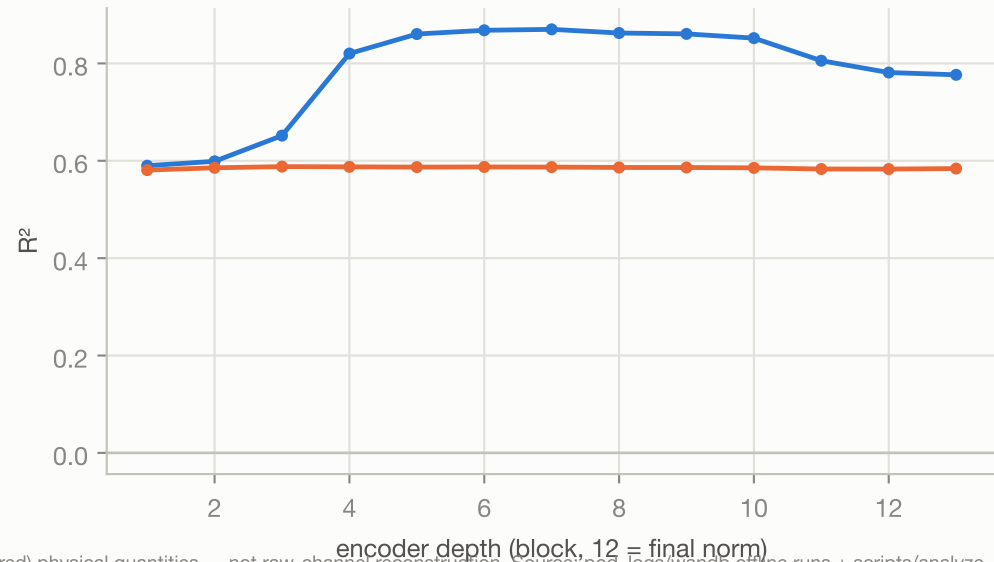
Gradient norm (training stability)



Physics decodability, final layer (ridge probe, 5-fold)



“convective flux” R² by depth



Loss curves EMA-smoothed ($\alpha=0.06$). R² = 5-fold ridge probe on mean-pooled encoder features vs. derived (non-stored) physical quantities — not raw-channel reconstruction. Source: pod_logs/wandb offline runs + scripts/analyze_encoder